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# CROWDING-CONTROLLED STREPTAVIDIN MOBILITY AND HYALURONAN BRUSH DIFFUSION ON SUPPORTED MEMBRANES

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## Abstract

Cell adhesion is mediated by soft cell-surface coatings such as the glycocalyx - a hydrated polysaccharide brush-like structure. It is thus modulated by softness, but also lateral mobility of the binding partners and multivalent ligand-receptor interactions (1). To quantitatively study the contribution of these parameters to adhesion strength, we use a minimal membrane platform in which molecular density, mobility and brush conformation can be varied independently. The platform consists of biotinylated supported lipid bilayers (SLBs), streptavidin (SAv) linkers (2) and biotinylated hyaluronan (b-HA) brushes. Using a combination of spectroscopic ellipsometry, confocal microscopy fluorescence recovery after photobleaching (FRAP), we first monitored streptavidin adsorption on biotinylated SLBs and lateral mobility of the resulting surface. We showed that by changing the incubation dose, SAv surface density could be tuned from dilute, mobile conditions to densely packed, for which crowding markedly slows down both adsorption and diffusion, suggesting that both processes are limited by the availability of free surface area. We then grafted b-HA chains onto the SAv layer, and monitored the thickness (and the related grafting density) using reflection interference contrast microscopy (RICM). FRAP measurements show that b-HA grafting further reduces the diffusion coefficient of HA chains, with an amplitude that depends on b-HA density and chain interactions. These measurements provide practical conditions to

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tune both the conformation of model HA brush (thickness, density and hence mechanical properties) and diffusion of chains within the brush. This system thus provides a versatile and well-controlled membrane platform to connect molecular mobility, soft-layer structure and multivalent adhesion.

**Keywords:** supported lipid bilayer, streptavidin, hyaluronan brush, molecular crowding, fluorescence recovery after photobleaching (FRAP), reflection interference contrast microscopy (RICM), multivalent adhesion